



DIGITAL LOGIC

CHAPTER 4: **FLIP-FLOP**

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Knowledge points in this chapter:

- **Positive logic and negative logic**
- **Common flip-flop functions, flip-flop modes and external working characteristics.**

4.0 Positive logic and negative logic

When discussing the logic functions of various logic gates, it is agreed to use a high level to indicate a logic 1 and a low level to indicate a logic 0. In fact, it can be specified that a logic 1 is represented by a high level, a logic 0 is represented by a low level, or logic 0 can be specified by a high level and a logic 1 by a low level. This leads to the concept of positive logic and negative logic.

Positive logic: logic 1 is represented by a high level and logic 0 is represented by a low level.

Negative logic: logic 0 is represented by a high level and logic 1 is represented by a low level.



All the various logic gates discussed earlier were defined in terms of positive logic.

In this module, if there is no special explanation, the problem is discussed in terms of positive logic. The symbols of all gate circuits are represented by positive logic.

4.1 Flip-Flop

In a digital system, in order to construct a logic circuit that implements various functions, in addition to logic gates that need to implement logic operations, we still require logic devices that can remember information. Flip-flop is an electronic device with a memory function.

A flip-flop can be used to store one bit of binary information. There are many types of integrated flip-flops, and the classification methods are also different, but in terms of its structure, they are all coupled by logic gates and appropriate feedback lines.

Flip-flop features :

☆ There are two outputs Q and $\overline{Q}$

☆ There are two steady/stable states. $Q = 1$ and $\overline{Q} = 0$ are usually called “**1**” status, and $Q = 0$ and $\overline{Q} = 1$ called “**0**” status. When the input signal does not change, the flip-flop state remains stable.

☆ Under certain input signal, the flip-flop can shift from one stable state to another stable state.

The concept of the present state and the next state :

Present state: The state before the input signal is applied, recorded as Q^n and $\overline{Q}^n$, Generally abbreviated as Q and $\overline{Q}$;

Next state: The state after the input signal is applied, recorded as Q^{n+1} and $\overline{Q}^{n+1}$.

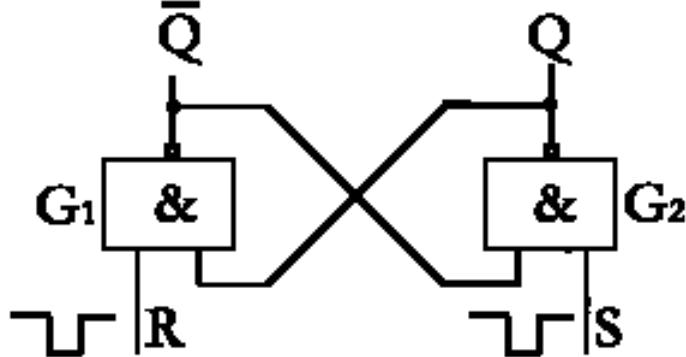
Obviously, the next state is a function of the present state and input.

4.1.2 Basic R-S flip-flop

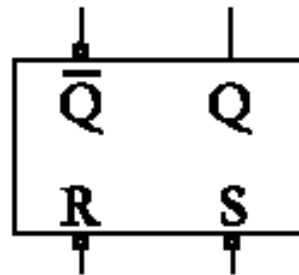
The basic R-S flip-flop is an abbreviation for direct reset and set flip-flop. Since it is a basic part that constitutes various function flip-flops, it is called a basic R-S flip-flop.

I. Basic R-S flip-flop constructed with NAND gates

1. Composition: It is composed of two NAND gates that are cross-coupled. The logical diagram and logical symbols are shown in Figures (a) and (b) below.



(a)

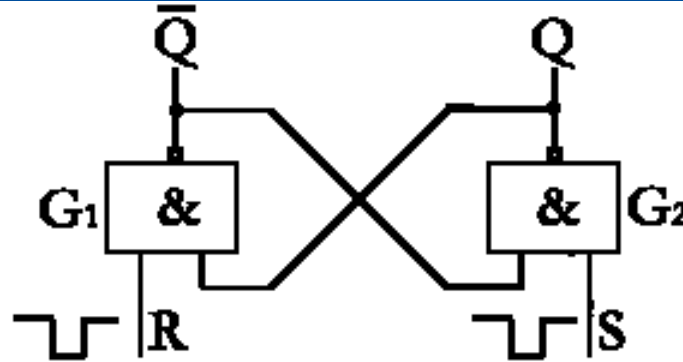


(b)

Basic R-S flip-flop constructed with NAND gates

In the picture, **R** is called a **0 port or reset port**. **S** is called a **1 port or set port**. A small circle added to the logic symbol input means that a low level or negative pulse is active.

2. Operating principle



- (1) If $R=1, S=1$, the flip-flop keeps its original state;
- (2) If $R = 1, S = 0$, the flip-flop is set to 1 state;
- (3) If $R = 0, S = 1$, the flip-flop is set to 0 state;
- (4) Do not allow $R=0, S=0$.

3. Logic function and its description

The logic function of the R-S flip-flop implemented by NAND gates is shown in the table below.

Basic R-S flip-flop function table

RS	Q^{n+1}	functional description
0 0	d	indefinite
0 1	0	set 0
1 0	1	set 1
1 1	Q	retain

The "d" in the table indicates that the next state of the flip-flop is undefined. This table is also called the next state truth table. (?!)

forbidden

reset

set

holding

RS	Q_n	Q_{n+1}
00	0	d
00	1	d
01	0	0
01	1	0
10	0	1
10	1	1
11	0	0
11	1	1

Next state truth table

Q_n	Q_{n+1}	R	S
0	0	d	1
0	1	1	0
1	0	0	1
1	1	1	d

**The excitation table of the basic
R-S flip-flop**

If the flip-flop Q^{n+1} is expressed as a function of the present state Q and input R, S , the Karnaugh map is as follows:

RS					
Q		00	01	11	10
	0	d	0	0	1
	1	d	0	1	1

The Next State Karnaugh Map of R-S Flip-Flop Constructed by NAND Gates

After simplifying the Karnaugh map, we can get the **characteristic equation** of this flip-flop:

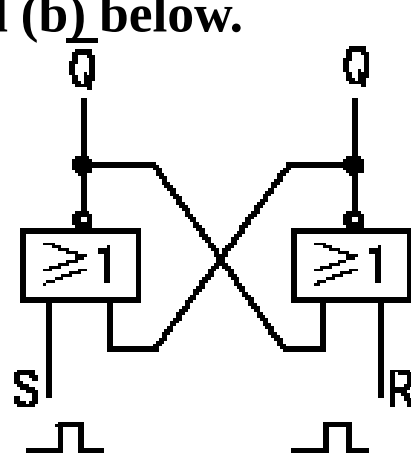
$$Q^{n+1} = \bar{S} + RQ$$

Because R and S do not allow 0 at the same time, the input must satisfy the **constraint**:

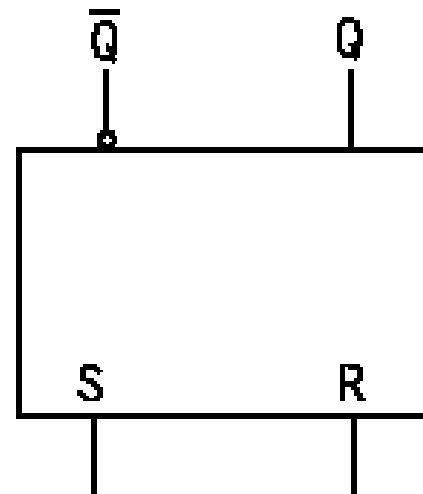
$$R + S = 1 \quad \text{(Constraint equation)}$$

II. Basic R-S flip-flop constructed with NOR gates

1. Composition : It is composed of two NOR gates that are cross-coupled. The logical diagram and logical symbols are shown in Figures (a) and (b) below.



(a)



(b)

Basic R-S flip-flop constructed with NOR gates

The input of the circuit is positive pulse or high level effective, so the input of the logic symbol does not add small circle.

2. Logic function

The following table shows the logical functions of R-S flip-flop constructed with NOR gates.

Basic R-S flip-flop function table The next state and constraint equations are as follows :

RS	Q^{n+1}	functional description
0 0	Q	constant
0 1	1	set 1
1 0	0	set 0
1 1	d	indefinite

$$Q^{n+1} = S + \overline{R}Q \quad \text{(next state equation)}$$

$$R \cdot S = 0 \quad \text{(constraint equation)}$$

The advantage of the basic R-S flip-flop is its **simple structure**. Not only can it be used independently as a memory element, but it is also used as an essential part of a variety of well-performed flip-flop because it has a direct reset and set function. **However, due to the constraint relationship between R and S, and the inability to perform timing control, its use is limited.**

4.1.3 Several common clock controlled flip-flop

The flip-flop with clock pulse controlled is called "**clock controlled flip-flop**" or "**timing flip-flop**".

The working characteristics of clock pulse controlled flip-flop:

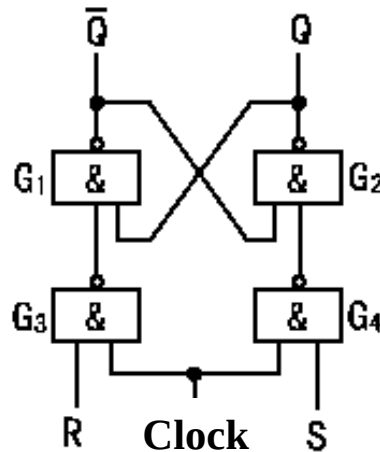
The moment of the state transition is determined by the clock pulse(When to transform?) ;

The input signal determines the direction of the flip-flop state transition(How to transform?) 。

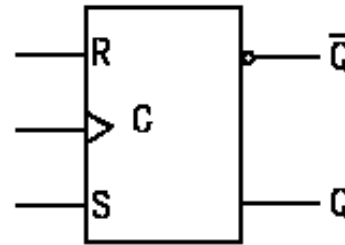
The four most commonly used clock controlled flip-flops are described below.

I. Clock Controlled/Synchronous R-S flip-flop

The logic diagram and logic symbols of the clock-controlled R-S flip-flop are shown in Figures (a) and (b).



(a)



(b)

1. Composition: Clock Controlled R-S flip-flop

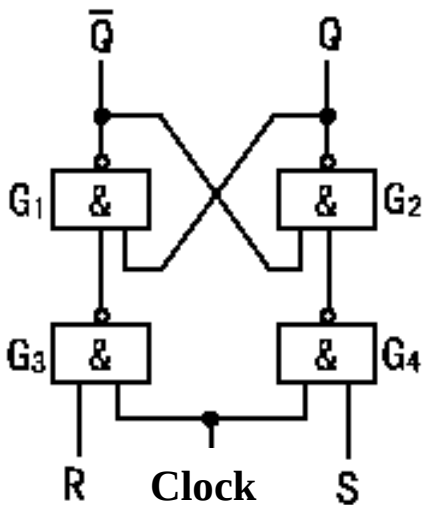
It consists of four NAND gates. Among them, the NAND gates G1 and G2 form a basic R-S flip-flop; the NAND gates G3 and G4 form a control circuit, which is usually called a control gate.

2. Operational principle

- When the clock pulse does not arrive (ie, $C = 0$), regardless of the values of the R and S port, the output of both control gates is 1 and the state of the flip-flop remains unchanged.

- When the clock pulse arrives (ie, $C = 1$), the values of the inputs R, S can be applied to the above basic R-S flip-flop through the control gate.

Details as follows:



R=0, S=0: The output of the control gate G3, G4 is 1 and the state of the flip-flop remains unchanged;

R=0, S=1: The outputs of the control gates G3 and G4 are 1 and 0, respectively, and the flip-flop state is set to 1;

R=1, S=0: The outputs of the control gates G3 and G4 are 0 and 1, respectively, and the flip-flop state is set to 0;

R=1, S=1: The outputs of the control gates G3 and G4 are all 0 and the flip-flop state is indefinite. This is not allowed.

The next-state truth table of synchronous R-S flip-flop

holding

set

reset

forbid

den

RS	Q_n	Q_{n+1}
00	0	0
00	1	1
01	0	1
01	1	1
10	0	0
10	1	0
11	0	d
11	1	d

Q_n	Q_{n+1}	R	S
0	0	d	0
0	1	0	1
1	0	1	0
1	1	0	d

The excitation
table of
synchronous R-S
flip-flop

It is known from the analysis that the working process of the clock controlled R-S flip-flop: it is enabled by the clock signal C and the input signal R and S; the clock C controls the conversion time and inputs the R and S to determine the state after the conversion.

3. Logic function

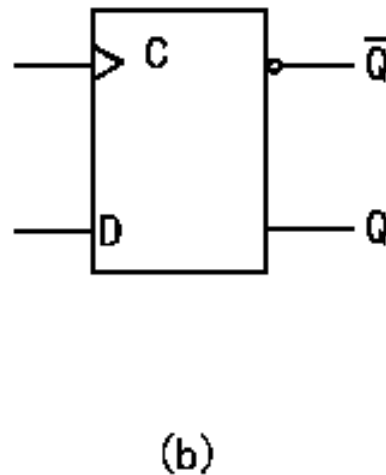
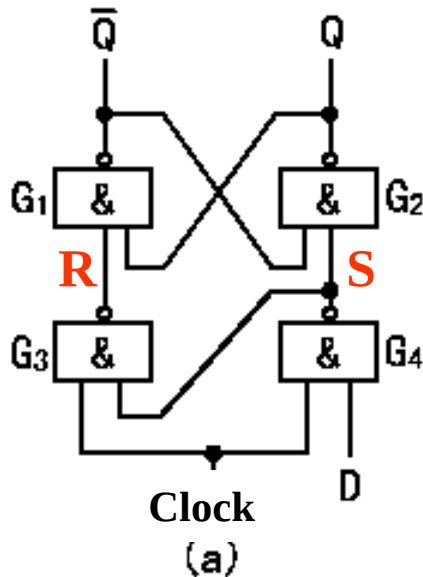
The function table, next state equations, and constraints of the clocked R-S flip-flop are the same as those of the R-S flip-flop formed by a NOR gate.

In the clock controlled flip-flop, Clock signal C is a fixed time reference and is not normally listed as an input signal in the table. When describing the function of the flip-flop, only consider the effect of the clock (C=1)**Note!** Although the clock control R-S flip-flop solves the problem of timing control of the flip-flop operation, and has the advantages of simple structure, there are still the following two limitations:

- The input signal still has constraints, that is, R and S cannot be 1 at the same time;

II. D flip-flop

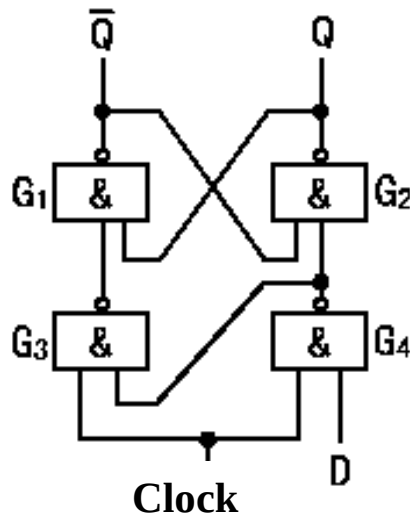
The control circuit of the clock-controlled R-S flip-flop is slightly modified to make it into the form shown in Figure (a) below, thus creating a D flip-flop with only one input. Its logical symbol is shown in Figure (b).



After modification, the control circuit converts the input signal D into a pair of complementary signals to the two input port of the basic R-S flip-flop during the clock pulse ($C=1$). It makes the two input signals of the basic R-S flip-flop can only be 01 or 10, thus eliminating the state uncertainty and solving the constraint on the input.

D flip-flop logic diagrams and logic symbols

Operating principle:



When there is no clock pulse (ie, $C=0$), the control circuit is blocked. Regardless of the value of D , the outputs of the NAND gates $G3$ and $G4$ are 1 and the state of the flip-flop remains unchanged.

When the clock pulse is applied (ie $C=1$), if $D=0$, the gate $G4$ output is 1, the gate $G3$ output is 0, the flip-flop state is set to 0; if $D=1$, the gate $G4$ output is 0, the gate $G3$ output is 1 and the flip-flop state is set to 1.

When the clock acts, the change of the state of the D flip-flop depends only on the input signal D , regardless of the present state.

The next state equation is

$$Q^{(n+1)} = D$$

The logic function of the D flip-flop is shown in the right table.

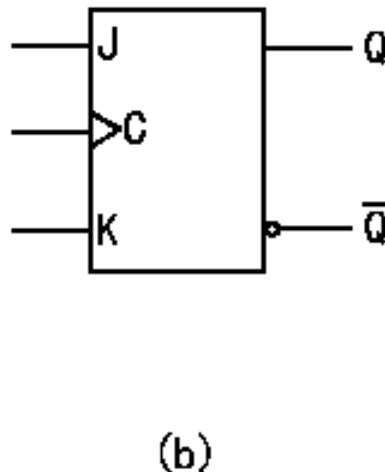
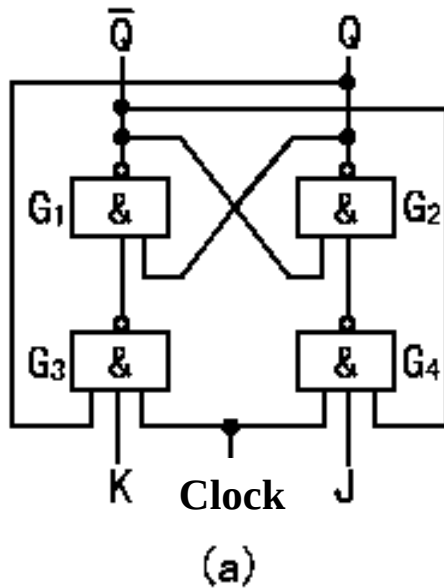
D flip-flop function table

D	Q^{n+1}
0	0
1	1

III. J-K flip-flop

Two feedback lines $\bar{Q}$ are added to the clock-controlled RS flip-flop, and the outputs Q and $\bar{Q}$ of the flip-flop are fed back to the input port of the two control gates, and the original input port S is changed to J, and R is changed to K, improved to JK flip-flops.

J-K flip-flop logic diagram and logic symbols as shown below:



The flip-flop utilizes the characteristics that the two output terminals of the flip-flop are always complementary to each other. It effectively solves the problem that the clock controlled R-S flip-flop can cause the flip-flop state to be uncertain when the two inputs are 1 at the time of clock pulse effecting.

J-K flip-flop logic diagrams and logic symbols

Operating principle:

(1) When there is no clock pulse ($C=0$), the flip-flop keeps original state.

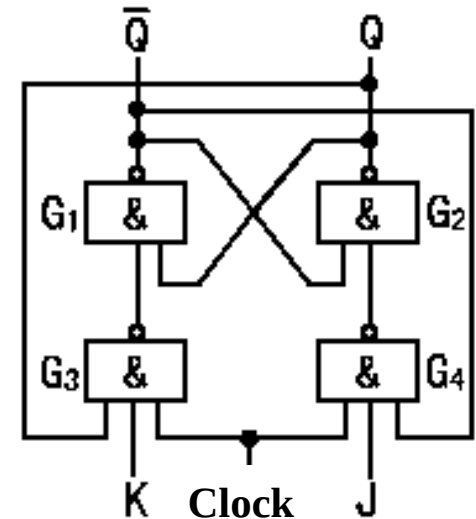
(2) When clock pulses ($C=1$), they are related to J and K.

① $J=0, K=0$: The flip-flop state is unchanged

② $J=0, K=1$: If it is in the 0 state, the flip-flop state remains at 0; if it is in the 1 state, the flip-flop state is set to 0. That is, $JK = 01$, the flip-flop state must be 0.

③ $J=1, K=0$: If it is in the 0 state, the flip-flop state is set to 1; if it is in the 1 state, the flip-flop state remains at 1. That is, $JK = 10$, the flip-flop state must be 1.

④ $J=1, K=1$: If it is in the 0 state, the flip-flop is set to the 1 state; if it is in the 1 state, the flip-flop is set to the 0 state. That is, when $JK = 11$, the next state of the flip-flop is opposite to the present state.



To sum up, the function table of the J-K flip-flop is shown in the following table.

J-K flip-flop function table

J K	Q^{n+1}	functional description
0 0	Q	constant
0 1	0	set 0
1 0	1	set 1
1 1	$\overline{Q}$	toggle

The above J-K flip-flop has a simple structure and strong logic function, but there is still a "over toggle" phenomenon. In order to further solve the "over toggle" problem, **master-slave J-K flip-flops** are widely used in practice.

The next-state truth table of synchronous J-K flip-flop

holding

reset

set

reverse

J K	Q_n	Q_{n+1}
0 0	0	0
0 0	1	1
0 1	0	0
0 1	1	0
1 0	0	1
1 0	1	1
1 1	0	1
1 1	1	0

The excitation

Q_n	Q_{n+1}	J	K
0	0	0	d
0	1	1	d
1	0	d	1
1	1	d	0

To sum up, the function table of the J-K flip-flop is shown in the following table.

J-K flip-flop function table

J K	Q^{n+1}	functional description
0 0	Q	constant
0 1	0	set 0
1 0	1	set 1
1 1	$\overline{Q}$	toggle

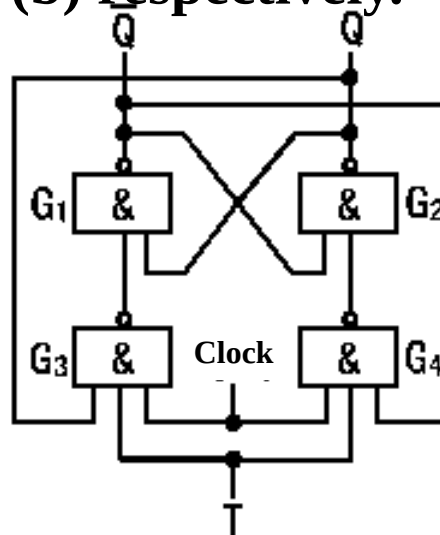
The next state equation is

$$Q^{n+1} = J\overline{Q} + \overline{K}Q$$

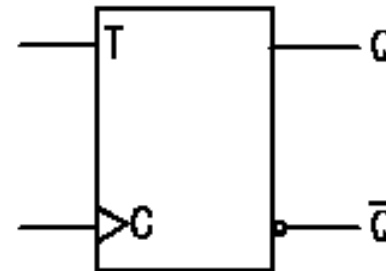
The above J-K flip-flop has a simple structure and strong logic function, but there is still a "over toggle" phenomenon. In order to further solve the "over toggle" problem, **master-slave J-K flip-flops** are widely used in practice.

IV. T flip-flop

T flip-flop, also known as counting flip-flop. If the two inputs J and K of the J-K flip-flop are connected and the connected inputs are denoted by the symbol T, the T flip-flop is formed. The corresponding logic diagrams and logic symbols are shown in (a) and (b) respectively.



(a)



(b)

T flip-flop logic diagrams and logic symbols

The logic function of the T flip-flop can be derived directly from the next state equation of the J-K flip-flop. The next state equation of the J-K flip-flop is

$$Q^{(n+1)} = J\bar{Q} + \bar{K}Q$$

After replacing J and K in this equation with T, the T-flip-flop's next state equation can be obtained:

$$Q^{(n+1)} = T\bar{Q} + \bar{T}Q$$

According to the next state equation, the function table of the T flip flop can be listed

T flip-flop function table

T	Q^{n+1}	functional description
0	$\bar{Q}$	constant
1	Q	toggle

Function:

**When T=0, the flip-flop state remains unchanged;
When T=1, the state toggles under the action of a clock pulse, which is equivalent to a binary counter.**

The flip-flop constituted by the four logic gates has a problem of toggling. The so-called “over toggle” refers to a phenomenon that the flip-flop state changes twice or more during the same clock pulse.

What is the cause of the “over toggle” ?

Because during the clock pulse, the input signal directly controls the change of the flip-flop state. That is, when the clock C is 1, the input signals R, S change, and the state of the flip-flop changes, causing multiple toggles during a clock pulse.



In order to overcome the "over toggle" phenomenon, integrated T-flip flops used in actual digital circuits generally adopt a master-slave structure or maintain a blocking structure. In addition to their performance improvements, the logic functions are exactly the same as those described above.



There are four different types of clock-controlled flip-flops described above, and logic functions can be converted between these flip-flops.

In general, adding a certain conversion logic circuit to the input of the original flip-flop can constitute a flip-flop with a new logic function. The conversion method can be found in the relevant section of the textbook.